

7 C

total gain in energy = total loss in energy

gain in GPE_P + gain in KE_{PQ} = loss in GPE_Q + W.D. by friction

$$m_P g (1.5 \sin 30^\circ - 0) + \text{gain in } KE_{PQ} = m_Q g (1.5 - 0) - f (1.5)$$

$$\begin{aligned} \text{gain in } KE_{PQ} &= m_Q g (1.5) - f (1.5) - m_P g (1.5 \sin 30^\circ) \\ &= (4.0)(9.81)(1.5) - (2.5)(1.5) - (3.0)(9.81)(1.5 \sin 30^\circ) \\ &= 33.0375 \\ &= 33 \text{ J} \end{aligned}$$

8 D

$$mv^2$$